SPX Fill Realism — August 2026

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INSTRUMENT SPX (SPXW) — 0DTE cash-settled index options	EXCHANGE Cboe
TRADING HOURS 09:30–16:00 ET (SPXW settles on the close)	BROKER Interactive Brokers
MODE SIMULATED — IB paper fills, checked vs real OPRA NBBO	DATE RANGE 2026-08-06 → 2026-09-04
TRADING DAYS 22	NBBO / PRINT SOURCE ThetaData (OPRA consolidated, tick)

In plain English

This checks whether our simulated option fills could really have happened. For each of our **518** buy/sell fills, we looked up the **real market** — the actual bid/ask and the real trades that printed — at the exact second the broker says we filled, and compared our price against it.

Bottom line: the fills look realistic — the same prices a real marketable order would get.

- **Where we filled.** The typical fill landed at position **0.0** between the bid and the ask (0 = the price you pay to cross the spread, 0.5 = the midpoint, 1 = the best possible price). A value near 0 is what a real market order looks like — **86.1%** of fills were at-or-worse than the midpoint, i.e. not too-good-to-be-true.
- **Was the size really there. 99.8%** of fills had at least one contract actually available at that price — so the fill wasn't a mirage; real depth was sitting there.
- **Did a real trade happen at our price.** Of the **407** fills we could check, **384** had a genuine single-leg trade print at our price (not a multi-leg package price). A fill without one is **not** impossible — see the note under the table.
- **Anything off. 2.1%** looked slightly better than the market — a sub-second timing artifact (the quote we captured was the last one just before the fill), not an impossible fill. **1** fill(s) had a too-old quote and were set aside.

P&L — sim vs real market, side by side

Book rebuilt from ONLY real data: entries + order-exits at the real marketable touch, expiries cash-settled at the official SPX 4pm close. **Both sides use IB's real commissions** (the sim's modeled $1.30/trade is corrected first), so the commissions row is Δ$0 and every remaining Δ is fills + settlement. 187 trades.

SIM BOOKED (MODELED FEES)		SIM @ REAL FEES		REAL-MARKET NET
$9,240	→	**$8,666**	→	**$8,160**

COMPONENT	SIM (REAL FEES)	REAL (TD)	Δ $	Δ %
`Entry premium collected`	$97,145	$96,710	−$435	-0.4%
`Exit + settlement`	−$87,662	−$87,732	−$70	-0.1%
`Gross P&L`	*$9,484*	*$8,978*	*−$506*	*-5.3%*
`Commissions (real IB, both sides)`	−$818	−$818	$0	+0.0%
`Net P&L (at real commissions)`	**$8,666**	**$8,160**	**−$506**	**-5.8%**

By exit type (both at real fees):

CLOSED	TRADES	SIM	REAL NET	Δ $
`order-closed`	72	−$12,164	−$12,619	−$455
`expired-closed`	115	$20,830	$20,780	−$50

REAL-MARKET NET P&L

$8,160

sim @ real fees $8,666

SIM OVERSTATES / TRADE

$3

-5.8% · fills/settlement only

FILL EVENTS

518

518 with a quote · 0 no logged price

SCORED FILLS

517

quote ok, fresh, price known

MEDIAN FILL POSITION

0.0

0=crossed · 0.5=mid · 1=touch

CONSERVATIVE (≤ MID)

86.1%

pos ≤ 0.5 — see the cumulative % column

SIZE ≥ 1 AT OUR PRICE

99.8%

≥ 3: 92.6% · ≥ 10: 83.0%

MEDIAN SIZE AT TOUCH

76

contracts resting at our price (max 1463)

SINGLE-LEG PRINT-CONFIRMED

384/407

cond 0/18 traded at our price

THROUGH THE BOOK

2.1%

pos > 1 (timing/data noise)

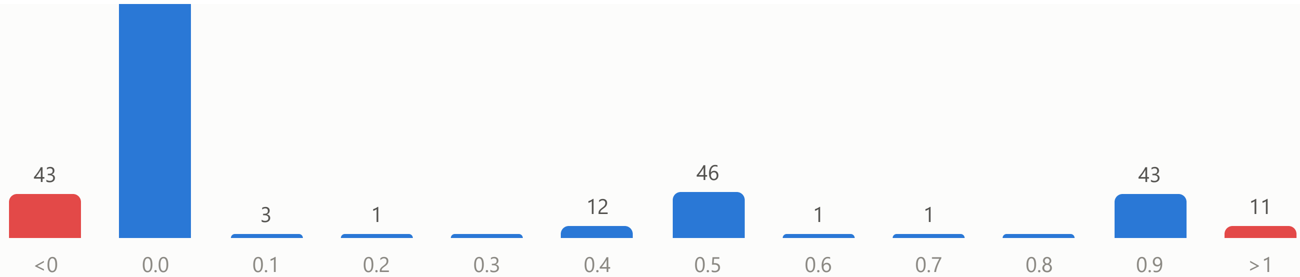
STALE QUOTES

1

quote >2s before fill (excluded)

Where our fills sat in the real NBBO

0 = crossed the spread (marketable, realistic) · 0.5 = mid · 1 = far touch (price improvement) · red = outside the book (timing/data noise). A real marketable engine clusters near 0.



Position breakdown — what share filled where

Every scored fill falls in exactly one band; the shares add to 100%. "Crossed the spread" is the realistic/conservative outcome.

WHERE OUR FILL LANDED	FILLS	% OF FILLS	CUMULATIVE %	MEDIAN POSITION
Crossed the spread — filled at the touch	399	77.2%	77.2%	0.000
Inside — better than the touch, at/below mid	46	8.9%	86.1%	0.500
Better than mid — price improvement	61	11.8%	97.9%	1.000
Through the book — sub-second timing noise	11	2.1%	100.0%	2.000
All scored fills	**517**	**100.0%**	**—**	**0.000**

By strategy

scored = quote ok, fresh, our price known. confirm% = single-leg prints (cond 0/18).

STRATEGY	EVENTS	SCORED	MEDIAN POSITION	CROSSED %	SIZE-OK %	CONFIRM %
eodfly_c	74	74	0.000	85%	100%	92%
eodfly_p	70	70	0.000	79%	100%	88%
eodic_c	32	32	0.000	94%	100%	100%
eodic_p	36	36	0.000	86%	100%	100%
gx_bcs	48	47	0.000	96%	100%	95%
gx_bps	44	44	0.000	82%	100%	100%
openfly_c	74	74	0.000	86%	100%	99%
openfly_p	78	78	0.000	79%	99%	89%
openic_c	22	22	0.000	91%	100%	100%
openic_p	40	40	0.000	98%	100%	100%

What "print-confirmed" means — and what it doesn't. A single-leg trade printing at our price is *positive* proof the fill was achievable. The reverse is not true: an unconfirmed fill is **not** an impossible one. This is paper, so our own order never prints to the tape — confirmation depends on some *other* trader printing at our exact price within ±3 seconds.

And spread legs frequently trade as multi-leg *package* orders (conditions 130/131/134), which we deliberately exclude. Those fills stand on the NBBO + size-at-touch check instead. In this run, 23 fills had single-leg prints nearby but not at our price, and a further set had no print in the window — neither is a red flag.

Method & FAQ

What this report covers — and what settles at the close

The 518 events here are **entry fills + order-based exit fills** — every time we actually traded against the market. That includes the "premise didn't hold" early exits and the few positions we closed with an order near the EOD. **Positions we held to expiry are not fills**: SPXW cash-settles at the official 4:00pm close, so those legs are validated separately against the settlement price, not the NBBO. That settlement carries a large share of the book's P&L, so it gets its own check — this report is the fill half.

How each fill is timed — and the lag

Every fill is anchored to **IB's true execution time** (Trade-Confirmation report, second precision). We also record our own fill time in the sim, but that is our machine's clock at the fill *callback*, which fires after the real execution. On the 47 fills we could match, our log lands a median **+1.00s** after IB's execution (mean +2.57s, 90th pct +10.40s; 83% within ±2s) — our clock+callback offset, which is exactly why we anchor the NBBO lookup on IB's exec time, not our clock.

What we don't yet have: the moment the sim *submitted* the order, so we can't quote a true submit→fill latency. That's one checkbox away — add "Order Time" to the IB Flex query and we compute it exactly.

The ±3-second window

The print-confirmation window is **centered on the fill**: 3s before to 3s after (6s total). Why 3s? A trade-off — too narrow (±1s) and thin 0DTE strikes often have no nearby single-leg print, so we'd under-confirm real fills; too wide (±9s) and on a fast tape you match prints from a *different* price regime and over-confirm. ±3s catches a genuinely contemporaneous trade without drifting. It affects only the *secondary* print check — the primary metric (where the fill sat in the NBBO) uses the exact quote at the fill instant and is **not** windowed. A volatile 3s is not a problem: more prints just means more chances to find a real trade at our price; we still only count one at/through it.

When would we say a fill could NOT have happened?

Two independent questions:

- **Could it happen? (primary)** At the fill instant, was our price at/inside the real NBBO *and* was size there? We capture the **actual** resting size, not just a flag — median **76 contracts** at our price (range 0–1463; only 0.2% had zero). A fill looks **not** achievable only when it printed **through the book** (better than the best available price — 2.1%) or had **zero size** — and the through-book cases are sub-second quote drift, not impossible fills.
- **Did a real trade print at our price? (secondary)** The 384/407 confirmation. Its absence — 23 had nearby single-leg prints but not at our price, others had none in the window — is **not** evidence against the fill; fillability is judged by the primary test.

What "position < 0" means

Position 0 = filled exactly at the marketable touch (buy at the ask / sell at the bid). **Below 0** means we filled *worse* than the visible touch — paid above the ask or sold below the bid — extra-conservative slippage where the quote moved against us. The **43** fills below 0 make the sim look *harder* on itself, not easier. **Above 1** is the reverse: better than the best price, the through-the-book timing tail.